Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	SWTID-2026-5113
Project Name	LegalEase
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent indentation used across the documented modules.
2	Proper File Structure	Files and folders are logically organized	Yes	Logical separation of frontend, backend, AI core and formatting modules.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Names such as document_type, parties, terms and dates are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as format_docx, format_pdf and format_html_preview are descriptive.
5	Code Comments	Inline and block comments explain logic	Partial	
6	Modular Design	Code is split into reusable functions/modules	Yes	
7	No Redundant Code	Duplicate or unused code is removed	Partial	No redundant-code cleanup is described explicitly in the project document.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	GeminiDocumentGenerator	Python / Gemini API	Legal document generation	High

2	format_docx	Python / python-docx	DOCX export workflow	High
3	format_pdf	Python / FPDF	PDF export workflow	High
4	format_html_preview	Python / Streamlit	Editable HTML preview	High
5	FastAPI /generate	Python / FastAPI	Frontend-to-AI document generation	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments	
Code Layout & Structure		4	Clear separation of Streamlit, FastAPI, AI integration and formatting modules.
Readability		4	Descriptive inputs and function names support readable maintenance.
Reusability		4	Reusable generation and formatting functions are explicitly defined.
Documentation / Comments		3	Project documentation is detailed; full inline-comment coverage is not evidenced.
Overall Score		4	Overall assessment based on the project documentation provided.